Customer Churn Risk Report

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Prediction

Likely to churn - churn probability **75.4%**

Risk level	High
Decision threshold	55.0% (above this = likely to churn)
Retention probability	24.6%

The model estimates this customer is at high risk of leaving. Contact them before renewal.

Customer details

Contract type	Month-to-month
Payment method	Electronic check
Internet service	Fiber optic
Online security	No
Tech support	No
Tenure	12 months
Monthly charges	$70.00
Total charges	$840.00

Key factors

Input	Value	Effect on estimate
Contract type	Month-to-month	Pushes the estimate up
Tenure	12 months	Pushes the estimate up
Payment method	Electronic check	Pushes the estimate up
Online security	No	Pushes the estimate up
Internet service	Fiber optic	Pushes the estimate up

Factors describe patterns the model associates with churn, not proven causes.

Recommended retention actions

• Offer a discount for moving to a one- or two-year contract.

• Invite the customer to an automatic payment method (bank transfer or card).

• Offer a trial of the online-security add-on.

• Ask about speed and reliability; fiber accounts churn more often in this data.

Model information

Model	Gradient Boosting
Evaluation	1409 held-out customers (stratified 20% test split, seed 42)
Accuracy / precision / recall	76.2% / 53.7% / 75.4%
F1-score	62.7%
ROC-AUC	0.838

PR-AUC (baseline = churn rate)	0.657 (baseline 0.265)
Brier score (lower is better)	0.164 (baseline 0.195)
Confusion matrix (TN/FP/FN/TP)	792 / 243 / 92 / 282
Data	Telco-Customer-Churn.csv (7,043 customers)

How to read the probability

• The predicted outcome compares the estimated probability with the decision threshold.

• Risk bands: Low below 40%, Moderate from 40% to below 70%, High from 70%.

• Key factors list the inputs that moved this customer's estimate the most.

Limitations

• The model was trained on one public telecom dataset (IBM Telco Customer Churn). Results may not transfer to other businesses, regions or time periods.

• The model does not see support history, usage, contract dates or competitor offers, which often matter for churn.

• Key factors show patterns the model associates with churn. They are not proven causes.

• Probabilities are estimates. Compare customers with each other rather than reading any single score as an exact frequency.

This report is generated by a statistical model trained on a public telecom dataset. The churn probability is an estimate of how similar customers behaved in that data. It is not a guarantee of what this customer will do, and it does not prove why they would leave. Use it together with your own knowledge of the customer.